

Interior Points Methods

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Why should I bother to learn this stuff?

- Interior point methods are competitive with simplex methods for linear programs
- Interior point methods are state of the art for most conic (convex) problems (inc. SOCP, SDP, etc.)
- $\implies$ useful for
 - ▶ understanding what is used in numerical solvers (like MOSEK, Gurobi, SCS, etc.)
 - ▶ specialization in optimization

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- 1 Recalls on convex differentiable optimization problems
- 2 Equality constrained optimization
- 3 Barrier methods [BV 11.2-11.3]
 - Interior penalization
 - Duality
 - Interpretation through KKT condition
- 4 Interior Point Method
- 5 Application to linear problem
- 6 Wrap-up

Convex differentiable optimization problem

We consider the following convex optimization problem

$$\begin{aligned} (\mathcal{P}) \quad & \min_{x \in \mathbb{R}^n} f(x) \\ & \text{s.t.} \quad Ax = b \\ & \quad \quad h_i(x) \leq 0 \quad \quad \forall i \in [n_I] \end{aligned}$$

where A is a $n_E \times n$ matrix, and all functions f and h_i are assumed convex, real-valued and twice differentiable.

$$\begin{aligned} (\mathcal{P}) \quad & \min_{x \in \mathbb{R}^n} f(x) \\ & \text{s.t.} \quad Ax = b \\ & \quad \quad h_i(x) \leq 0 \quad \forall i \in [n_I] \end{aligned}$$

is equivalent to

$$\min_{x \in \mathbb{R}^n} f(x) + \mathbb{I}_{\{Ax=b\}} + \sum_{i=1}^{n_I} \mathbb{I}_{\{h_i(x) \leq 0\}}$$

which we rewrite

$$\min_{x \in \mathbb{R}^n} f(x) + \sup_{\lambda \in \mathbb{R}^{n_E}} \lambda^\top (Ax - b) + \sum_{i=1}^{n_I} \sup_{\mu_i \geq 0} \mu_i h_i(x)$$

$$\begin{aligned} (\mathcal{P}) \quad & \min_{x \in \mathbb{R}^n} f(x) \\ & \text{s.t.} \quad Ax = b \\ & \quad \quad h_i(x) \leq 0 \quad \forall i \in [n_I] \end{aligned}$$

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which we rewrite

$$\min_{x \in \mathbb{R}^n} \sup_{\lambda \in \mathbb{R}^{n_E}, \mu \in \mathbb{R}_+^{n_I}} f(x) + \lambda^\top (Ax - b) + \sum_{i=1}^{n_I} \mu_i h_i(x)$$

$$(\mathcal{P}) \quad \min_{x \in \mathbb{R}^n} \sup_{\lambda \in \mathbb{R}^{n_E}, \mu \in \mathbb{R}_+^{n_I}} \underbrace{f(x) + \lambda^\top (Ax - b) + \sum_{i=1}^{n_I} \mu_i h_i(x)}_{:= \mathcal{L}(x; \lambda, \mu)}$$

$$(\mathcal{D}) \quad \sup_{\lambda \in \mathbb{R}^{n_E}, \mu \in \mathbb{R}_+^{n_I}} \min_{x \in \mathbb{R}^n} f(x) + \lambda^\top (Ax - b) + \sum_{i=1}^{n_I} \mu_i h_i(x)$$

As for any function ϕ we always have

$$\sup_y \inf_x \phi(x, y) \leq \inf_x \sup_y \phi(x, y)$$

we have that (weak duality)

$$\text{val}(\mathcal{D}) \leq \text{val}(\mathcal{P}).$$

$$(\mathcal{P}) \quad \min_{\mathbf{x} \in \mathbb{R}^n} \sup_{\lambda \in \mathbb{R}^{n_E}, \mu \in \mathbb{R}_+^{n_I}} \underbrace{f(\mathbf{x}) + \lambda^\top (A\mathbf{x} - b) + \sum_{i=1}^{n_I} \mu_i h_i(\mathbf{x})}_{:= \mathcal{L}(\mathbf{x}; \lambda, \mu)}$$

$$(\mathcal{D}) \quad \sup_{\lambda \in \mathbb{R}^{n_E}, \mu \in \mathbb{R}_+^{n_I}} \min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) + \lambda^\top (A\mathbf{x} - b) + \sum_{i=1}^{n_I} \mu_i h_i(\mathbf{x})$$

As for any function ϕ we always have

$$\sup_y \inf_x \phi(\mathbf{x}, \mathbf{y}) \leq \inf_x \sup_y \phi(\mathbf{x}, \mathbf{y})$$

we have that (weak duality)

$$\text{val}(\mathcal{D}) \leq \text{val}(\mathcal{P}).$$



Define the dual function

$$d(\lambda, \mu) := \inf_x \mathcal{L}(x; \lambda, \mu)$$

Then we have $val(\mathcal{D}) = \sup_{\lambda \in \mathbb{R}^{n_E}, \mu \in \mathbb{R}_+^{n_I}} d(\lambda, \mu)$.

Thus, we can compute a lower bound to $val(\mathcal{D}) \leq val(\mathcal{P})$ by choosing any admissible dual points $\lambda \in \mathbb{R}^{n_E}, \mu \in \mathbb{R}_+^{n_I}$ and solving the unconstrained problem

$$d(\lambda, \mu) = \inf_{x \in \mathbb{R}^n} f(x) + \lambda^\top (Ax - b) + \sum_{i=1}^{n_I} \mu_i h_i(x)$$



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$$d(\lambda, \mu) = \inf_{x \in \mathbb{R}^n} f(x) + \lambda^\top (Ax - b) + \sum_{i=1}^{n_I} \mu_i h_i(x)$$

Constraint qualification

Recall that, for a **convex** differentiable optimization problem, the constraints are qualified if *Slater's condition* is satisfied :

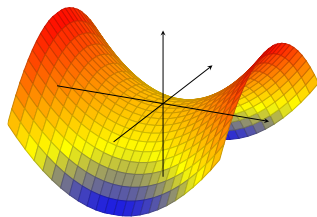
$$\exists x_0 \in \mathbb{R}^n, \quad Ax_0 = b, \quad \forall i \in [n_I], \quad h_i(x_0) < 0$$

i.e., there exists a strictly admissible feasible point



If $(\mathcal{P})$ is a convex optimization problem with qualified constraints, then

- $val(\mathcal{D}) = val(\mathcal{P})$
- any optimal solution $x^\#$ of $(\mathcal{P})$ is part of a saddle point $(x^\#; \lambda^\#, \mu^\#)$ of $\mathcal{L}$
- $(\lambda^\#, \mu^\#)$ is an optimal solution of $(\mathcal{D})$





If Slater's condition is satisfied, then $x^\#$ is an optimal solution to (P) if and only if there exists optimal multipliers $\lambda^\# \in \mathbb{R}^{n_E}$ and $\mu^\# \in \mathbb{R}^{n_I}$ satisfying

$$\begin{cases} \nabla f(x^\#) + A^\top \lambda^\# + \sum_{i=1}^{n_I} \mu_i^\# \nabla h_i(x^\#) = 0 & \text{first order condition} \\ Ax^\# = b & \text{primal admissibility} \\ h(x^\#) \leq 0 \\ \mu^\# \geq 0 & \text{dual admissibility} \\ \mu_i^\# h_i(x^\#) = 0, \quad \forall i \in [n_I] & \text{complementarity} \end{cases}$$

The three last conditions are sometimes compactly written

$$0 \geq h(x^\#) \perp \mu \geq 0$$



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Intuition for Newton's method: unconstrained case



Newton's method is an iterative optimization method that minimizes a quadratic approximation of the objective function at the current point $x^{(k)}$.

Consider the following unconstrained optimization problem (f smooth):

$$\min_{x \in \mathbb{R}^n} f(x)$$

At $x^{(k)}$ we have

$$f(x^{(k)} + d) = f(x^{(k)}) + \nabla f(x^{(k)})^\top d + \frac{1}{2} d^\top \nabla^2 f(x^{(k)}) d + o(\|d\|^2)$$

And the direction $d^{(k)}$ minimizing the quadratic approximation is given by solving for d

$$\nabla f(x^{(k)}) + \nabla^2 f(x^{(k)}) d = 0.$$

Intuition for Newton's method: constrained case



Approximate the linearly constrained optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & Ax = b \end{aligned}$$

by

$$\begin{aligned} \min_{d \in \mathbb{R}^n} \quad & f(x^{(k)}) + \nabla f(x^{(k)})^\top d + \frac{1}{2} d^\top \nabla^2 f(x^{(k)}) d \\ \text{s.t.} \quad & A(x^{(k)} + d) = b \end{aligned}$$

Which is equivalent to solving (for given admissible $x^{(k)}$)

$$\begin{aligned} \min_{d \in \mathbb{R}^n} \quad & \nabla f(x^{(k)})^\top d + \frac{1}{2} d^\top \nabla^2 f(x^{(k)}) d \\ \text{s.t.} \quad & Ad = 0 \end{aligned}$$

Finding Newton's direction

$$\begin{aligned} \min_{d \in \mathbb{R}^n} \quad & \nabla f(x^{(k)})^\top d + \frac{1}{2} d^\top \nabla^2 f(x^{(k)}) d \\ \text{s.t.} \quad & Ad = 0 \end{aligned}$$

By KKT the optimal $d^{(k)}$ is given by solving for (d, λ)

$$\begin{cases} \nabla f(x^{(k)}) + \nabla^2 f(x^{(k)})d + A^\top \lambda = 0 \\ Ad = 0 \end{cases}$$

Or in matrix form

$$\begin{pmatrix} \nabla^2 f(x^{(k)}) & A^\top \\ A & 0 \end{pmatrix} \begin{pmatrix} d \\ \lambda \end{pmatrix} = \begin{pmatrix} -\nabla f(x^{(k)}) \\ 0 \end{pmatrix}$$

Finding Newton's direction

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Newton's algorithm: equality constrained case

Data: Initial admissible point x_0

Result: quasi-optimal point

$k = 0$;

while $\|\nabla f(x^{(k)}) + A^\top \lambda\| \geq \varepsilon$ **do**

 Solve for d

$$\begin{pmatrix} \nabla^2 f(x^{(k)}) & A^\top \\ A & 0 \end{pmatrix} \begin{pmatrix} d \\ \lambda \end{pmatrix} = \begin{pmatrix} -\nabla f(x^{(k)}) \\ 0 \end{pmatrix}$$

 Line-search for $\alpha \in [0, 1]$ on $f(x^{(k)} + \alpha d^{(k)})$

$$x^{(k+1)} = x^{(k)} + \alpha d^{(k)}$$

$k \leftarrow k + 1$

Algorithm 1: Newton's algorithm

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Video explanation

A short video introduction to the content of this and the next section.

<https://www.youtube.com/watch?v=MsgpS15JRbI>

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Constrained optimization problem

We now want to consider a convex differentiable optimization problem with equality and inequality constraints.

$$\begin{aligned} (\mathcal{P}_\infty) \quad & \min_{x \in \mathbb{R}^n} f(x) \\ & \text{s.t. } Ax = b \\ & h_i(x) \leq 0 \quad \forall i \in [n_I] \end{aligned}$$

where all functions f and h_i are assumed convex, finite valued and twice differentiable.

Which we rewrite

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) + \sum_{i=1}^{n_I} \mathbb{I}_{\mathbb{R}^-}(h_i(x)) \\ \text{s.t.} \quad & Ax = b \end{aligned}$$

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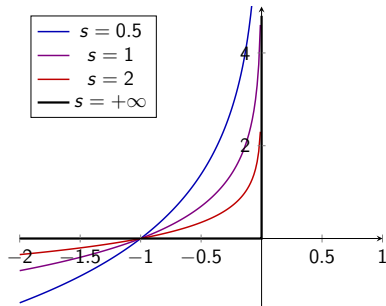
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The negative log function

- The idea of barrier methods is to replace the indicator function $\mathbb{I}_{\mathbb{R}^-}$ by a smooth function.
- We choose the function $z \mapsto -1/s \log(-z)$
- Note that this function also takes value $+\infty$ on $\mathbb{R}^+$
- ➔ In particular these barrier functions ensure **strict** feasibility of the iterates.





- We define

$$\phi(x) := \begin{cases} -\sum_{i=1}^{n_I} \ln(-h_i(x)), & \text{if } h_i(x) < 0 \ \forall i \in [n_I], \\ +\infty, & \text{otherwise.} \end{cases}$$

- Thus we have $\frac{1}{s}\phi(x) \xrightarrow{s \rightarrow +\infty} \mathbb{I}_{\{h_i(x) < 0, \forall i \in [n_I]\}}$
- On the domain $\{x : h_i(x) < 0\}$, we have:

$$\nabla \phi(x) =$$

$$\nabla^2 \phi(x) =$$



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- On the domain $\{x : h_i(x) < 0\}$, we have:

$$\nabla \phi(x) = \sum_{i=1}^{n_I} -\frac{1}{h_i(x)} \nabla h_i(x)$$

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$$\phi(\mathbf{x}) := \begin{cases} -\sum_{i=1}^{n_I} \ln(-h_i(\mathbf{x})), & \text{if } h_i(\mathbf{x}) < 0 \ \forall i \in [n_I], \\ +\infty, & \text{otherwise.} \end{cases}$$

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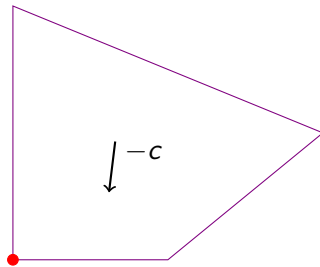
$$\begin{aligned} \nabla \phi(\mathbf{x}) &= \sum_{i=1}^{n_I} -\frac{1}{h_i(\mathbf{x})} \nabla h_i(\mathbf{x}) \\ \nabla^2 \phi(\mathbf{x}) &= \sum_{i=1}^{n_I} \left[\frac{1}{h_i^2(\mathbf{x})} \nabla h_i(\mathbf{x}) \nabla h_i(\mathbf{x})^\top - \frac{1}{h_i(\mathbf{x})} \nabla^2 h_i(\mathbf{x}) \right] \end{aligned}$$



We consider

$$\begin{aligned} (\mathcal{P}_\infty) \quad & \min_{x \in \mathbb{R}^n} f(x) \\ \text{s.t.} \quad & Ax = b \end{aligned}$$

with optimal solution $x^\sharp$.



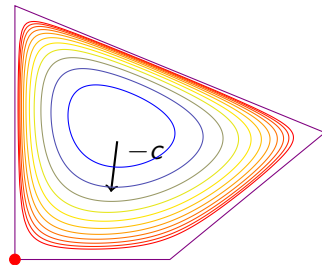


We consider

$$\begin{aligned} (\mathcal{P}_s) \quad & \min_{x \in \mathbb{R}^n} f(x) + \frac{1}{s} \phi(x) \\ \text{s.t.} \quad & Ax = b \end{aligned}$$

with optimal solution x_s .

scaling objective function by s does not change the solution.



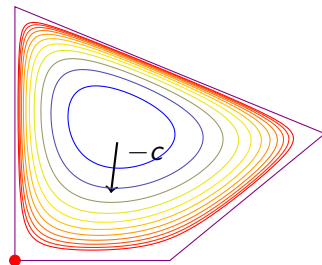


We consider

$$\begin{aligned} (\mathcal{P}_s) \quad & \min_{x \in \mathbb{R}^n} s f(x) + \phi(x) \\ & \text{s.t. } Ax = b \end{aligned}$$

with optimal solution x_s .

Letting s go to $+\infty$ makes x_s go to the solution of $(\mathcal{P})$ along the **central path**.



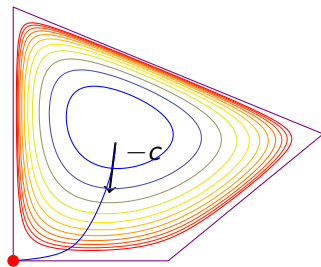


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x_s is solution of

$$\begin{aligned} (\mathcal{P}_s) \quad & \min_{x \in \mathbb{R}^n} s f(x) + \phi(x) \\ \text{s.t.} \quad & Ax = b \end{aligned}$$

if and only if, there exists $\lambda_s \in \mathbb{R}^{n_E}$, such that

x_s is solution of

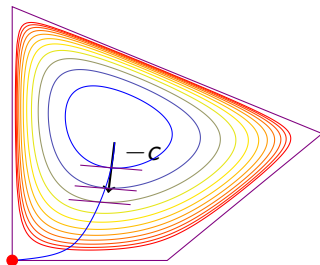
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if and only if, there exists $\lambda_s \in \mathbb{R}^{n_E}$, such that

$$\begin{cases} Ax_s = b \\ h_i(x_s) < 0 \\ s \nabla f(x_s) + \nabla \phi(x_s) + A^\top \lambda = 0 \end{cases} \quad \forall i \in [n_I]$$

$$\begin{cases} Ax_s = b \\ h(x_s) < 0 \\ \textcolor{brown}{s} \nabla f(x_s) + \nabla \phi(x_s) + A^T \textcolor{violet}{\lambda} = 0 \end{cases}$$

If $A = 0$ it means that $\nabla f(x_s)$ is orthogonal to the level lines of ϕ



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Duality



Recall the original optimization problem

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with Lagrangian

$$\mathcal{L}(x; \lambda, \mu) := f(x) + \lambda^\top (Ax - b) + \sum_{i=1}^{n_I} \mu_i h_i(x)$$

and dual function

$$d(\lambda, \mu) := \inf_{x \in \mathbb{R}^n} \mathcal{L}(x; \lambda, \mu).$$

For any admissible dual point $(\lambda, \mu) \in \mathbb{R}^{n_E} \times \mathbb{R}_+^{n_I}$, we have

$$d(\lambda, \mu) \leq \text{val}(\mathcal{P}_\infty)$$

Duality



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Getting a lower bound

For given admissible dual point $(\lambda, \mu) \in \mathbb{R}^{n_E} \times \mathbb{R}_+^{n_I}$, a point $x^\sharp(\lambda, \mu)$ minimizing $\mathcal{L}(\cdot, \lambda, \mu)$, is characterized by first order conditions

$$\nabla f(x^\sharp(\lambda, \mu)) + A^\top \lambda + \sum_{i=1}^{n_I} \mu_i \nabla h_i(x^\sharp(\lambda, \mu)) = 0$$

which gives

$$d(\lambda, \mu) = \mathcal{L}(x^\sharp(\lambda, \mu); \lambda, \mu) \leq \text{val}(\mathcal{P}_\infty)$$

Dual point on the central path



Now recall that x_s , solution of $(\mathcal{P}_s)$, is characterized by

$$\begin{cases} Ax_s = b, h(x_s) < 0 \\ s \nabla f(x_s) + \nabla \phi(x_s) + A^\top \lambda_s = 0 \end{cases}$$

And we have seen that

$$\nabla \phi(x) = \sum_{i=1}^{n_I} \frac{1}{-h_i(x)} \nabla h_i(x)$$

Thus,

$$\nabla f(x_s) + A^\top \lambda_s / s + \sum_{i=1}^{n_I} \underbrace{\frac{1}{-s h_i(x_s)}}_{(\mu_s)_i} \nabla h_i(x_s) = 0$$

which means that $x_s = x^\sharp(\lambda_s / s, \mu_s)$.

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which means that $x_s = x^\sharp(\lambda_s / s, \mu_s)$.

Bounding the error



Let x_s be a primal point on the central path satisfying

$$\exists \lambda_s \in \mathbb{R}^{n_E}, \quad s \nabla f(x_s) + \nabla \phi(x_s) + A^\top \lambda_s = 0$$

We define a dual point $(\mu_s)_i = \frac{1}{-s h_i(x_s)} > 0$. We have

$$\begin{aligned} d(\lambda_s/s, \mu_s) &= \mathcal{L}(x_s; \lambda_s/s, \mu_s) \\ &= f(x_s) + \frac{1}{s} \lambda_s^\top \underbrace{(A x_s - b)}_{=0} + \sum_{i=1}^{n_I} \frac{1}{-s h_i(x_s)} h_i(x_s) \\ &= f(x_s) - \frac{n_I}{s} \leq \text{val}(\mathcal{P}_\infty) \end{aligned}$$

➡ x_s is an n_I/s -optimal solution of $(\mathcal{P}_\infty)$.

Bounding the error



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A point x_s is on the central path iff it is strictly admissible and there exists $\lambda \in \mathbb{R}^{n_E}$ such that

$$\nabla f(x_s) + A^\top \lambda + \sum_{i=1}^{n_I} \underbrace{\frac{1}{-sh_i(x_s)}}_{(\mu_s)_i} \nabla h_i(x_s) = 0$$

which can be rewritten

$$\begin{cases} \nabla f(x_s) + A^\top \lambda + \sum_{i=1}^{n_I} (\mu_s)_i \nabla h_i(x_s) = 0 \\ Ax_s = b, h_i(x_s) < 0 \\ \mu_s > 0 \\ -(\mu_s)_i h_i(x_s) = \frac{1}{s} \end{cases} \quad \forall i \in [n_I]$$

Contents

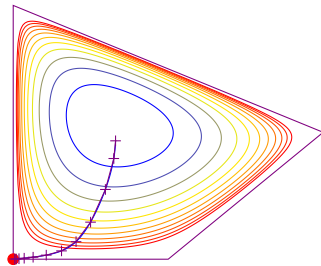
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- 2 Equality constrained optimization
- 3 Barrier methods [BV 11.2-11.3]
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- We saw that we can extend Newton's method to solve linearly constrained optimization problem.
- We saw that we can approximate inequality constraints through the use of logarithmic barrier $-1/s \sum_i \ln(-h_i(x))$.
- We proved that x_s is an n_I/s -optimal solution.
- The trade-off with s is: larger s means x_s closer to optimal solution x_∞ but the approximate problem $(\mathcal{P}_s)$ has worse conditioning.



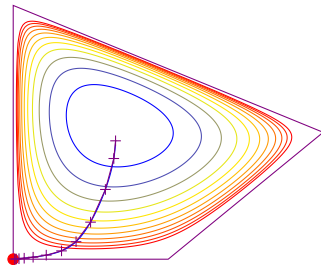
Data: increase $\rho > 1$, error $\varepsilon > 0$,
initial s
Result: ε -optimal point
solve $(\mathcal{P}_s)$ and set $x = x_s$;
while $n_I/s \geq \varepsilon$ **do**
 increase s : $s = \rho s$;
 centering step: solve $(\mathcal{P}_s)$ starting
 at x ;
 update : $x = x_s$





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Question : why solve $(\mathcal{P}_s)$ to optimality ?



Solving $(\mathcal{P}_s)$ with Newton's method

$$\begin{aligned} (\mathcal{P}_s) \quad & \min_{x \in \mathbb{R}^n} \quad sf(x) + \phi(x) \\ & \text{s.t.} \quad Ax = b \end{aligned}$$

is a linearly constrained optimization problem that can be solved by Newton's method. More precisely we have $x^{(k+1)} = x^{(k)} + d^{(k)}$ with $d^{(k)}$ a solution of

$$\begin{pmatrix} s\nabla^2 f(x^{(k)}) + \nabla^2 \phi(x^{(k)}) & A^\top \\ A & 0 \end{pmatrix} \begin{pmatrix} d \\ \lambda \end{pmatrix} = \begin{pmatrix} -s\nabla f(x^{(k)}) - \nabla \phi(x^{(k)}) \\ 0 \end{pmatrix}$$

Path following interior point method

Data: increase $\rho > 1$, error $\varepsilon > 0$, initial s_0

initial strictly feasible point x_0

$k = 0$

$x \leftarrow x_0$, $s \leftarrow s_0$

for $k \in \mathbb{N}$ **do**

// Outer step

for $\kappa \in [K]$ **do**

// Inner step

 solve for d ;

// Newton step for $(\mathcal{P}_s)$

$$\begin{pmatrix} s\nabla^2 f(x) + \nabla^2 \phi(x) & A^\top \\ A & 0 \end{pmatrix} \begin{pmatrix} d \\ \lambda \end{pmatrix} = \begin{pmatrix} -s\nabla f(x) - \nabla \phi(x) \\ 0 \end{pmatrix}$$

 reduce α from 1 until $x + \alpha d$ is strictly feasible and

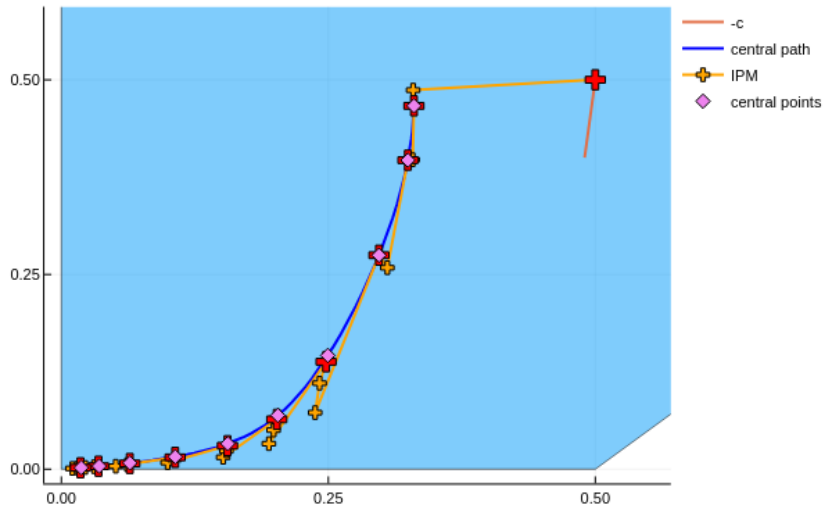
$$sf(x + \alpha d) + \phi(x + \alpha d) \leq sf(x) + \phi(x);$$

$x \leftarrow x + \alpha d$;

$s \leftarrow \rho s$;

Algorithm 2: Path following algorithm

Path following algorithm



Video explanation

A longer presentation to watch at a later time

<https://www.youtube.com/watch?v=zm4mfr-QT1E>

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A linear problem - inequality form

We consider the following LP

$$\begin{array}{ll} \min_{x \in \mathbb{R}^n} & c^\top x \\ \text{s.t.} & a_i^\top x \leq b_i \quad \forall i \in [n_I] \end{array}$$

Where $a_i^\top = A[i, :]$ is the row of matrix A , such that the constraints can be written $Ax \leq b$.
Thus, x_s is the solution of

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where

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Thus, x_s is the solution of

$$\min_{x \in \mathbb{R}^n} \quad sc^\top x + \phi(x)$$

where

$$\phi(x) := - \sum_{i=1}^{n_I} \ln(b_i - a_i^\top x)$$



$$\phi(\mathbf{x}) = - \sum_{i=1}^{n_I} \ln(b_i - \mathbf{a}_i^\top \mathbf{x})$$

$$\nabla \phi(\mathbf{x}) =$$

$$\nabla^2 \phi(\mathbf{x}) =$$



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$$\nabla \phi(\mathbf{x}) = \sum_{i=1}^{n_I} \frac{1}{b_i - \mathbf{a}_i^\top \mathbf{x}} \mathbf{a}_i$$

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This can be written in matrix form, using the vector $\mathbf{d} \in \mathbb{R}^{n_I}$ defined by $d_i = \frac{1}{b_i - \mathbf{a}_i^\top \mathbf{x}}$

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$$\nabla \phi(\mathbf{x}) = \mathbf{A}^\top \mathbf{d}$$

$$\nabla^2 \phi(\mathbf{x}) = \mathbf{A}^\top \text{diag}(\mathbf{d})^2 \mathbf{A}$$



Starting from x , the Newton direction for $(\mathcal{P}_s)$ is

$$dir_s(x) =$$

which, in algebraic form, yields

$$dir_s(x) =$$

with $d_i = 1/(b_i - a_i^\top x)$.



Starting from x , the Newton direction for $(\mathcal{P}_s)$ is

$$dir_s(x) = -(\nabla^2 \phi(x))^{-1}(sc + \nabla \phi(x))$$

which, in algebraic form, yields

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$$dir_s(x) = -[A^\top \text{diag}(d)^2 A]^{-1}(sc + A^\top d)$$

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Theory tells us to use a step-size of 1 for Newton's method.



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with $d_i = 1/(b_i - a_i^\top x)$.

Theory tells us to use a step-size of 1 for Newton's method.

Practice teaches us to use a smaller step-size (or line search).

Interior Point Method for LP pseudo code

Data: Initial admissible point x_0 , initial penalization $s_0 > 0$;

parameter : $\rho > 1$, $N_{in} \geq 1$, $N_{out} \geq 1$;

Result: quasi-optimal point

$x = x_0$, $s = s_0$;

for $k = 1..N_{out}$ **do**

for $\kappa = 1..N_{in}$ **do**

 Compute d , with $d_i = 1/(b_i - a_i^T x)$;

 Solve for dir

$$A^T \text{diag}(d)^2 A \text{dir} = -(sc + A^T d)$$

 reduce α from 1 until $b - A(x + \alpha \text{dir}) > 0$ and

$sc^T(x + \alpha \text{dir}) + \phi(x + \alpha \text{dir}) \leq sc^T x + \phi(x)$;

 update $x \leftarrow x + \alpha \text{dir}$;

 update $s \leftarrow \rho s$;

Algorithm 3: Interior Point Method for LP

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What you have to know

- IPMs are state-of-the-art algorithms for LPs and, more generally, conic optimization problems
- Logarithmic barriers are useful inner penalization methods

What you really should know

- That Newton's algorithm can be applied with equality constraints
- What the central path is
- That IPMs work with inner and outer optimization loops